The data structure of an API response, for example, will be referenced within integration tests to check for successful responses. Integration environments rely on a database with data for specific requests. Therefore, the assertions will be made against the specific data that was set up prior to the test running. However, contract tests focus on the structure of the data, providing matches for checking value types. Therefore, any data could be returned from the API response only to verify the structure of the data. The difference between the checks highlights that contract tests verify the structure of the data, and integration tests verify the actual data that is returned, as shown in figure 12.3.

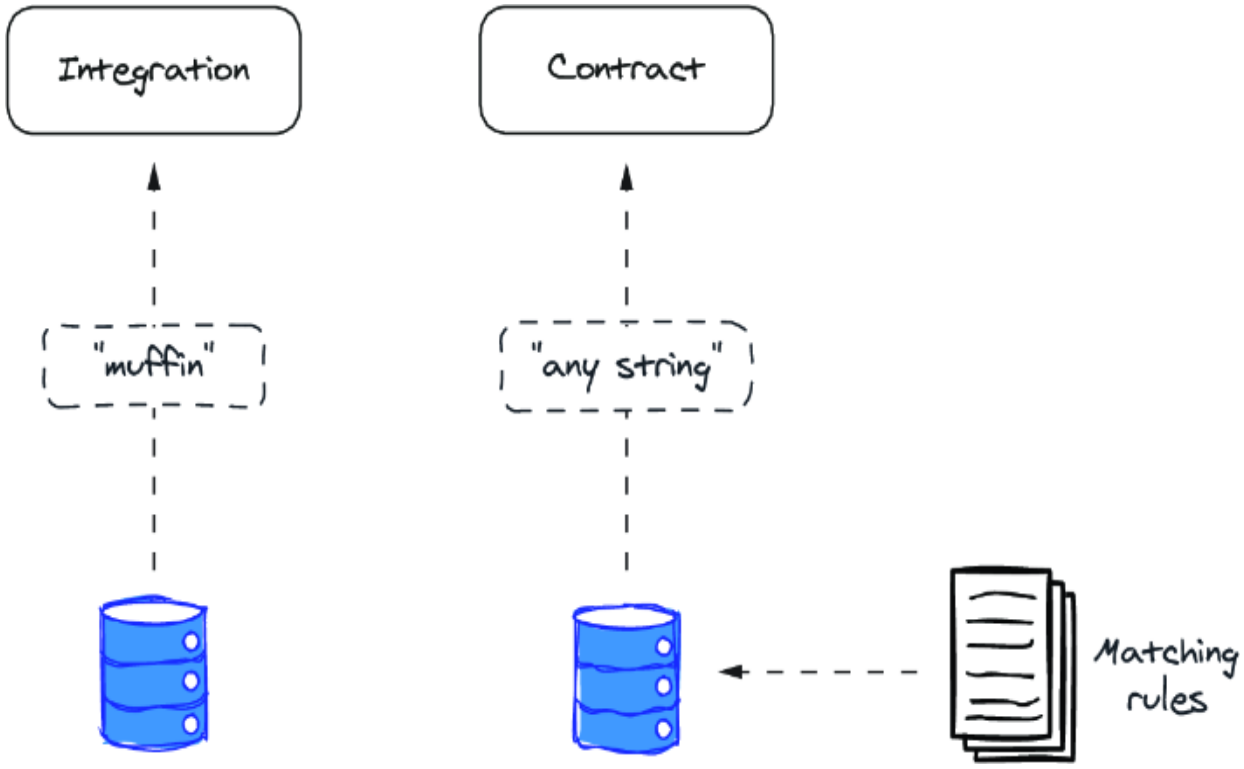


Figure 12.3 Integration tests return exact data, and contract tests return a version of the data type defined within the contract to make assertions more flexible.

The HTTP status codes associated with the request reflect the possible scenarios that occur from the response. For example, with integration tests and contract tests, scenarios need to be set to configure HTTP error conditions. The difference here is that consumers share their expectations of error conditions with the provider service. Usually, integration tests live alongside the provider code, therefore not fully understanding the consumer's expectations. With contract tests being run against local